

JACK J. BURLESON

AI / MACHINE LEARNING • DATA SCIENCE

Madison, WI • Open to Relocation • 262-341-1561 • jackjburleson@proton.me • linkedin.com/in/jackjburleson

PROFESSIONAL SUMMARY

UW-Madison M.S. candidate in Psychology: Data Science in Human Behavior (expected Dec 2026); B.S. in Psychology (neuroscience concentration, minor in sociology), UW-Parkside. Applied AI practitioner building full-lifecycle systems spanning multi-agent document-intelligence pipelines, transformer fine-tuning and calibration, LLM evaluation, computer vision classification, and behavioral prediction models. Master's capstone with American Family Insurance acting as industry partner, targeting methods to decrease costs of AI inference & improve document intelligence workflows. I bring five years of high-volume logistical & systems operations experience in quality assurance, documentation, and iterative process improvement.

TECHNICAL SKILLS

Programming Languages: Python (pandas, NumPy, scikit-learn, PyTorch, Hugging Face), R, MATLAB, SQL, Parquet, Git

Machine Learning: Transformer fine-tuning, model evaluation & calibration, cross-validation, SHAP/feature importance, error analysis, computer vision (OCR-based document classification), experimental design

AI Engineering & Development: LangGraph, multi-agent pipelines, prompt engineering, LLM eval harnesses & LLM-as-a-judge, model serving (vLLM, Ollama, LM Studio, OpenRouter), ONNX + FastAPI deployment on Modal

Engineering & Cloud: Docker, pytest, Quarto, AWS, Azure

PROJECTS

Digital-Mailroom | *DSHB Capstone project — American Family Insurance (industry partner)*

- A capstone project with American Family Insurance targeting document intelligence and processing improvements; 10 Python packages, including entity extraction, classification, routing, and scalable workflows, to improve document intelligence and management efficiency.
- Designed and implemented a unified LangGraph pipeline integrating legal-document routing, prompt experimentation, deterministic scoring, dataset management, and deployment workflows in a reproducible Docker environment.

Mailroom-ML | *ML Training, Evaluation & Fine-Tuning*

- Developed a ModernBERT document classifier with end-to-end training, evaluation, and fine-tuning pipelines; data lineage tracking, probability calibration, and production routing logic included.
- Implemented PyTorch training pipelines with Transformers and Hugging Face datasets, scipy-based evaluation, and ONNX export for FastAPI serving on Modal; maintained pytest coverage throughout.

RVL-CDIP-Classifer | *Computer Vision & Prompt Engineering*

- Developed a Python toolkit that converts TIFF/PDF pages to 300-DPI images, runs OCR, and classifies documents into 16 RVL-CDIP categories with OpenRouter vision models.
- Ran 17+ prompt iterations and generated confusion matrices, misclassification analyses, and cost projections to guide evidence-based prompt engineering and model selection.

Psychometric Digital Twin | *Behavioral ML Research | UW-Madison*

- Built a behavioral prediction pipeline on 252 matched survey participants: traditional ML models (Ridge, Elastic Net, k-NN, Random Forest, XGBoost, MLP) predicting communication-apprehension scores from demographic, employment, geographic, and transit features, with cross-validation and SHAP feature importance.
- Validated the PRCA's psychometric structure with factor analysis, then built LLM "digital twin" personas via tiered embodiment prompts (Openrouter/Vllm) and benchmarked them against the ML baselines on shared metrics (MAE, band accuracy, F1), analyzing systematic bias by demographic group.

EDUCATION

University of Wisconsin–Madison | Madison, WI | Expected Graduation: December 2026

M.S. Candidate — Psychology: Data Science in Human Behavior | GPA: 3.85

- Graduate work in machine learning, statistics, behavioral data analysis, predictive modeling, experimental design, and publication-quality visualization.

University of Wisconsin–Parkside | Kenosha, WI | September 2020 – May 2025

B.S., Psychology | Concentration: Neuroscience | GPA: 3.93

- Minor in Sociology; Provost's List (7x), Dean's List (8x), Distinguished Achievement Award (2025), and Psychological Research Certification (300 lab hours).

PROFESSIONAL EXPERIENCE

UW–Parkside — Undergraduate Researcher, Cognitive Neuroscience Lab | Kenosha, WI | Jun 2024 - Aug 2025

- Conducted cognitive neuroscience research under Dr. Melissa Gregg, designing and pilot-testing an experimental study involving perceptions of chatbot truth, deception, and user cognitive workload (study code: github.com/Exios66/truth-deception-architecture).
- Operated EEG systems & analyzed neural time-series data to quantify brain activity, contributing signal-processing and statistical insights to preexisting lab research on sleep quality and cognitive load.
- Presented findings at the 2025 Midwestern Psychological Association conference.

UW–Madison Recreation & Wellbeing — Building Services | Madison, WI | Jan 2026–Present

- Maintain recreation and wellbeing centers across campus to high standards of cleanliness, safety, and operational readiness.
- Respond to facility and equipment issues across multiple locations, keeping centers safe and fully operational during peak use.
- Coordinated with operations and event staff to resolve facility issues quickly and minimize disruption for students, staff, and visitors.

Avis Budget Group — Driver | Madison, WI | Jun 2026–Present

- Transport, stage, and safety-check rental vehicles across a 500+ vehicle fleet, ensuring readiness and on-time availability across the rental network.
- Coordinate vehicle logistics and movements between locations while upholding fleet safety standards.

Meijer Distribution — Logistics Coordinator | Pleasant Prairie, WI | Jul 2020–Dec 2025

- Promoted from Retail Loader to Logistics Coordinator for reliability in quality assurance and data-driven process documentation across high-volume operations.
- Analyzed quality-assurance data for 64 daily outbound trailers across the Midwest regional network, maintaining error rates below 1% through root-cause investigation and process refinement.
- Partnered with warehouse and transportation teams to investigate quality problems and resolve logistics exceptions before shipment.

RESEARCH TRAINING

CITI Program: Conflicts of Interest training [2025]

CITI Program: IRB Training for Students and Faculty [2025]